O4 Conceptual Diagrams

iNHCES System Architecture, ERD, DFDs, Pipeline Flow, User Journey

TETFund NRF 2025 | Department of Quantity Surveying, ABU Zaria

Objective:	O4 - Conceptual Models
Content:	Six publication-quality diagrams for all O4 design artefacts
Diagrams:	Architecture | ERD | DFD Level 0 | DFD Level 1 | Pipeline | User Journey
Date:	26 April 2026
Purpose:	Visual companion to O4_01 through O4_04 documents
Note:	All diagrams are AMBER - AI-authored from O3 SRS and O4 design decisions

DATA SOURCE DECLARATION -- READ BEFORE USE

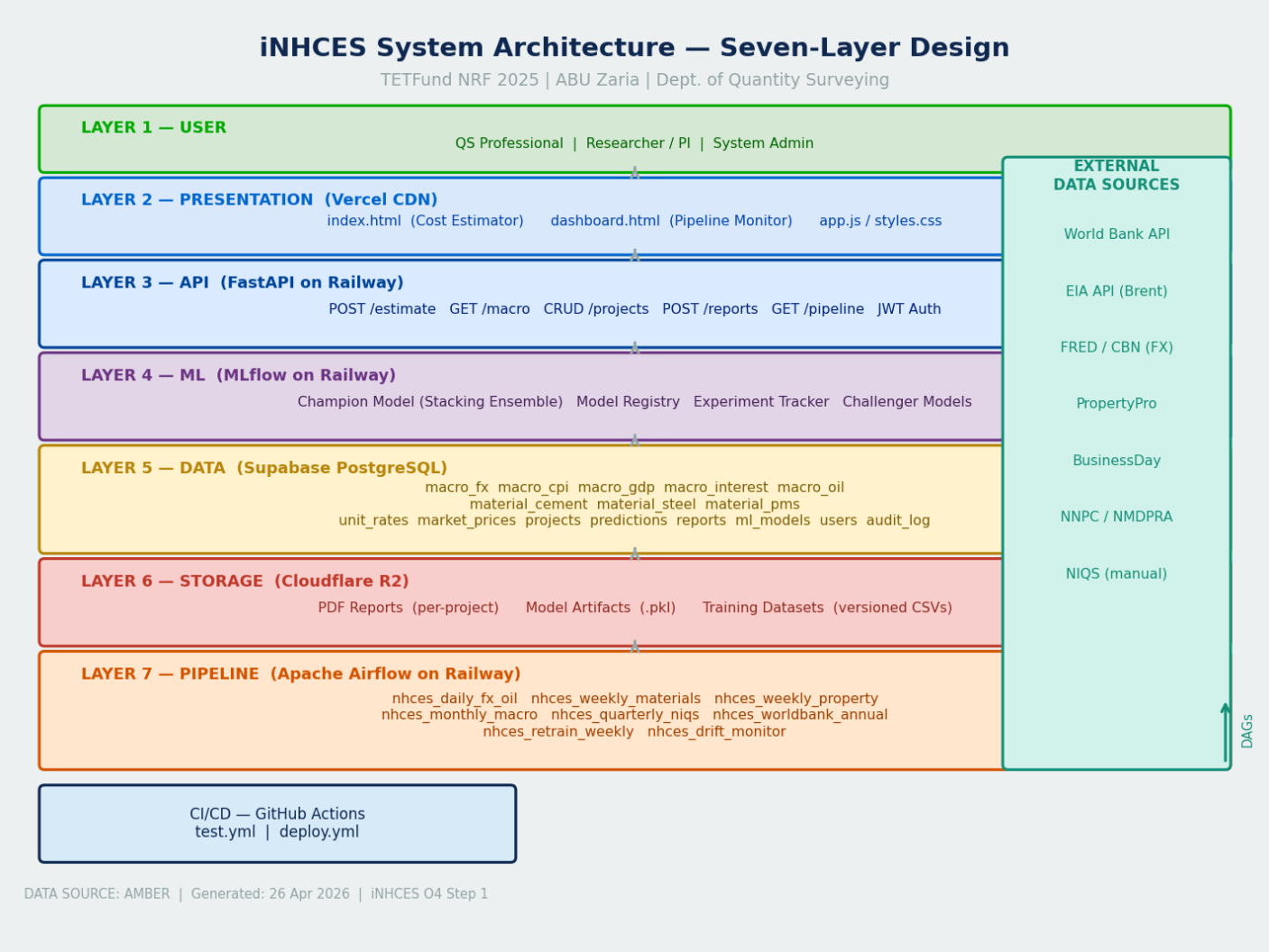
DATA SOURCE: AMBER -- AI-authored conceptual diagrams. Validate all diagrams against O3 SRS before O6 build.

All six diagrams were generated programmatically (matplotlib) by Claude Code from the iNHCES design specifications in O4 Steps 1-3. They represent intended system design, not implemented behaviour.

Render the Mermaid .mmd files in mermaid.live for alternative views of the same concepts. These PNG diagrams are optimised for PDF embedding and paper figures.

Diagram 1 - System Architecture

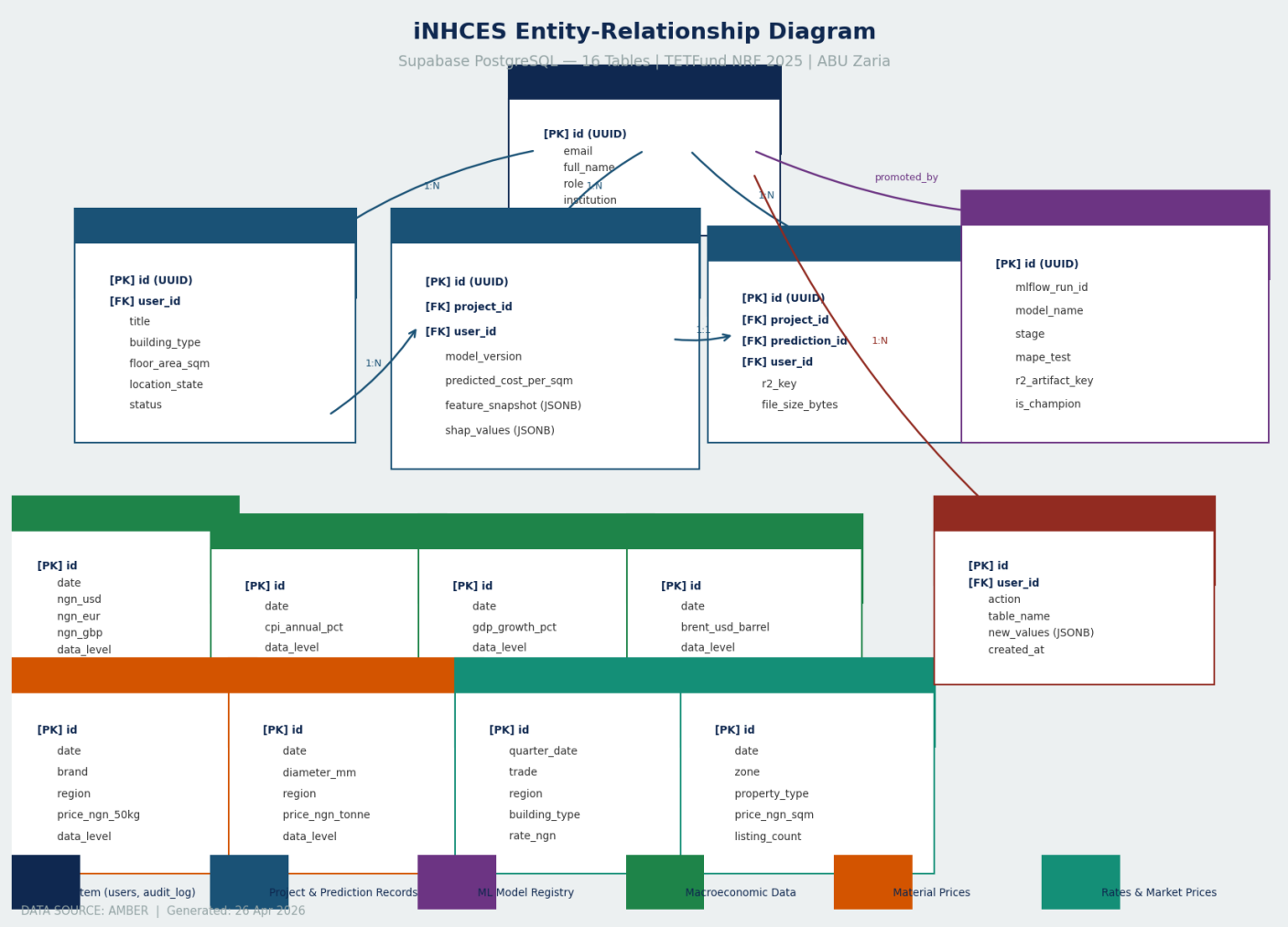
Seven-layer architecture showing all components, hosting platforms, and data flow directions between the User, Presentation, API, ML, Data, Storage, and Pipeline layers.



DATA SOURCE: AMBER | diag_01_architecture.png | iNHCES O4 | ABU Zaria / TETFund NRF 2025

Diagram 2 - Entity-Relationship Diagram (ERD)

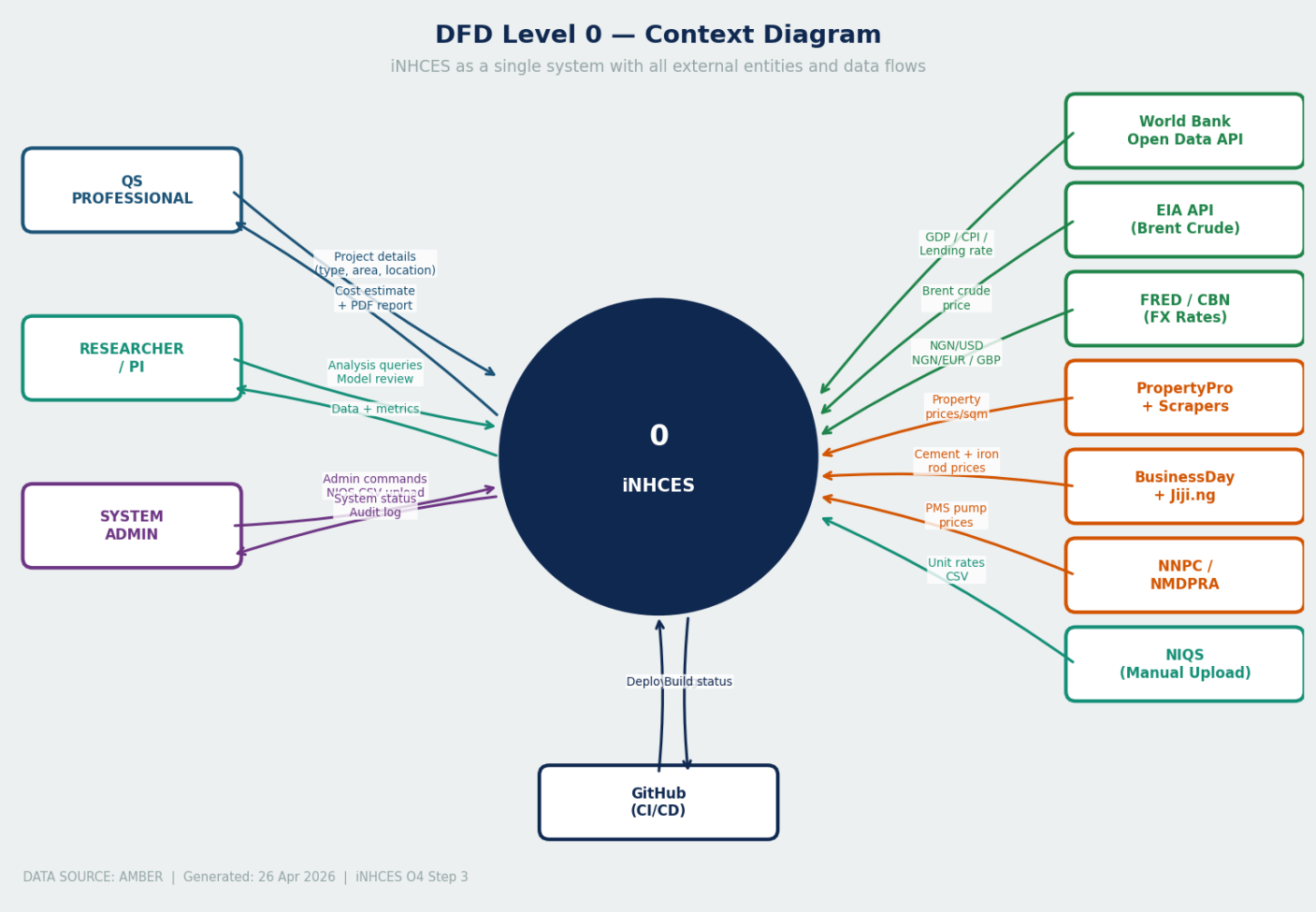
All 16 Supabase PostgreSQL tables with primary keys (PK), foreign keys (FK), key columns, and 1:N / 1:1 relationship lines. Colour-coded by functional group.



DATA SOURCE: AMBER | diag_02_erd.png | iNHCES O4 | ABU Zaria / TETFund NRF 2025

Diagram 3 - DFD Level 0 (Context Diagram)

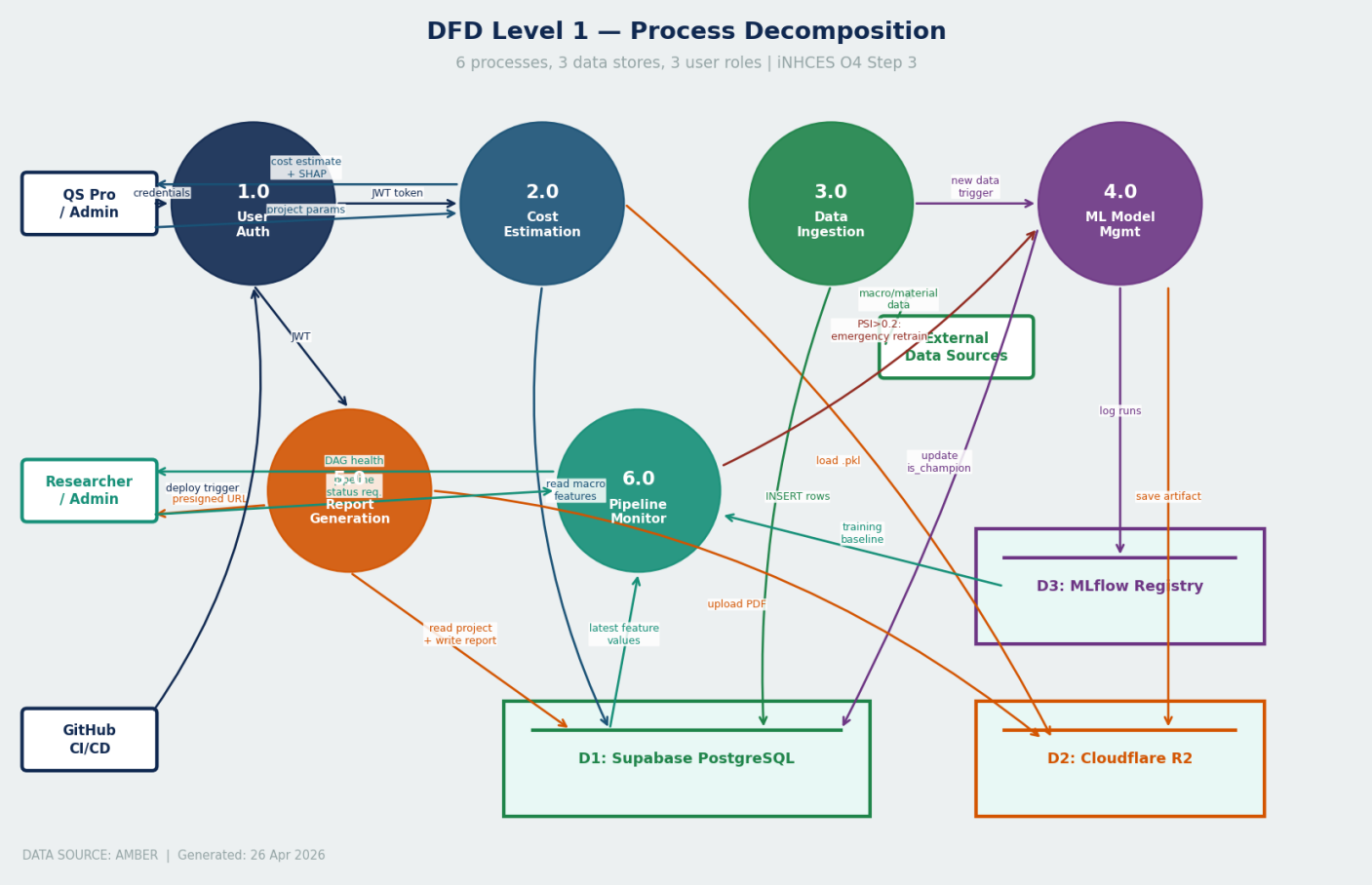
iNHCES as a single system bubble (Process 0) with all 11 external entities (3 user roles + 7 data sources + GitHub CI/CD) and all inbound/outbound data flows labelled.



DATA SOURCE: AMBER | diag_03_dfd_level0.png | iNHCES O4 | ABU Zaria / TETFund NRF 2025

Diagram 4 - DFD Level 1 (Process Decomposition)

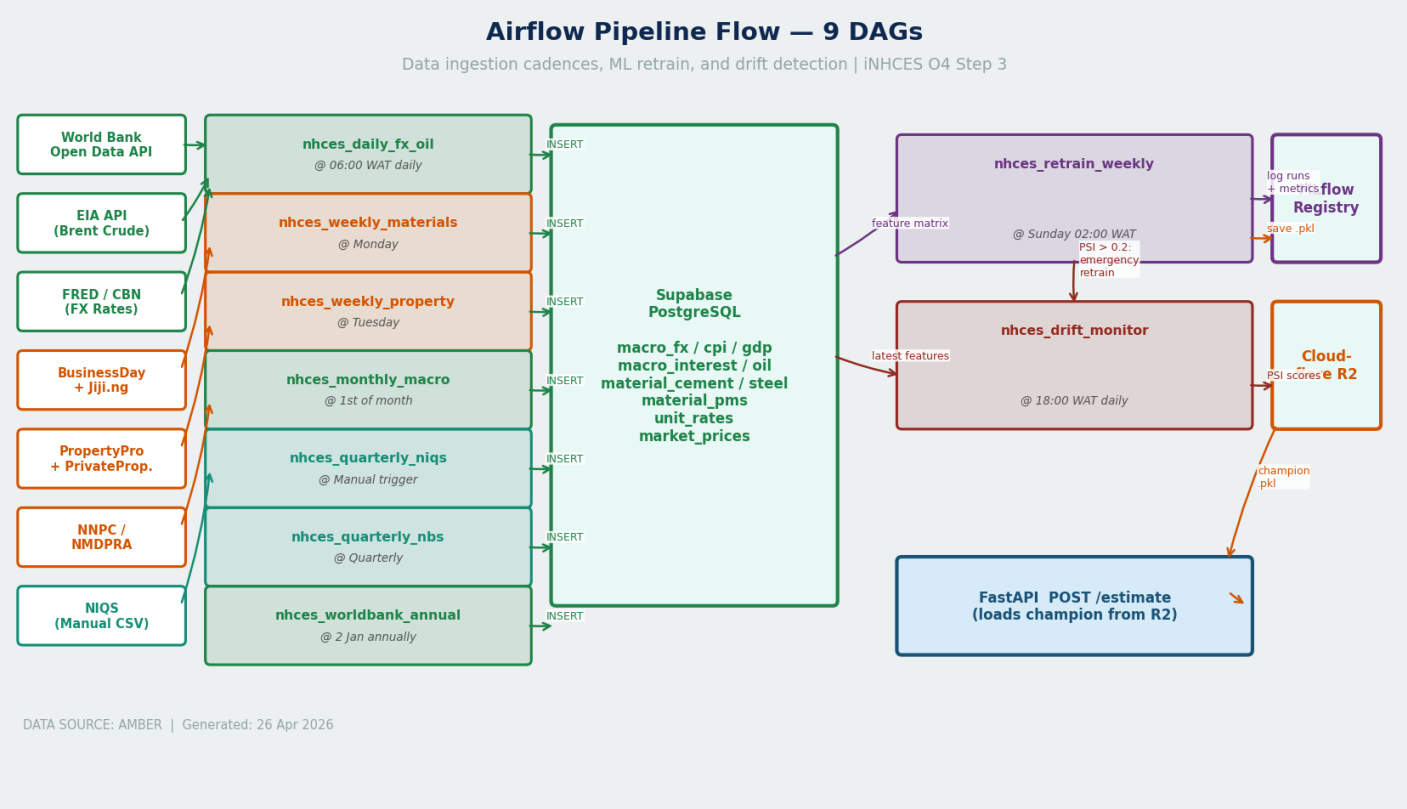
System decomposed into 6 processes (Auth, Cost Estimation, Data Ingestion, ML Management, Report Generation, Pipeline Monitor), 3 data stores (Supabase, R2, MLflow), and all inter-process data flows.



DATA SOURCE: AMBER | diag_04_dfd_level1.png | iNHCES O4 | ABU Zaria / TETFund NRF 2025

Diagram 5 - Airflow Pipeline Flow

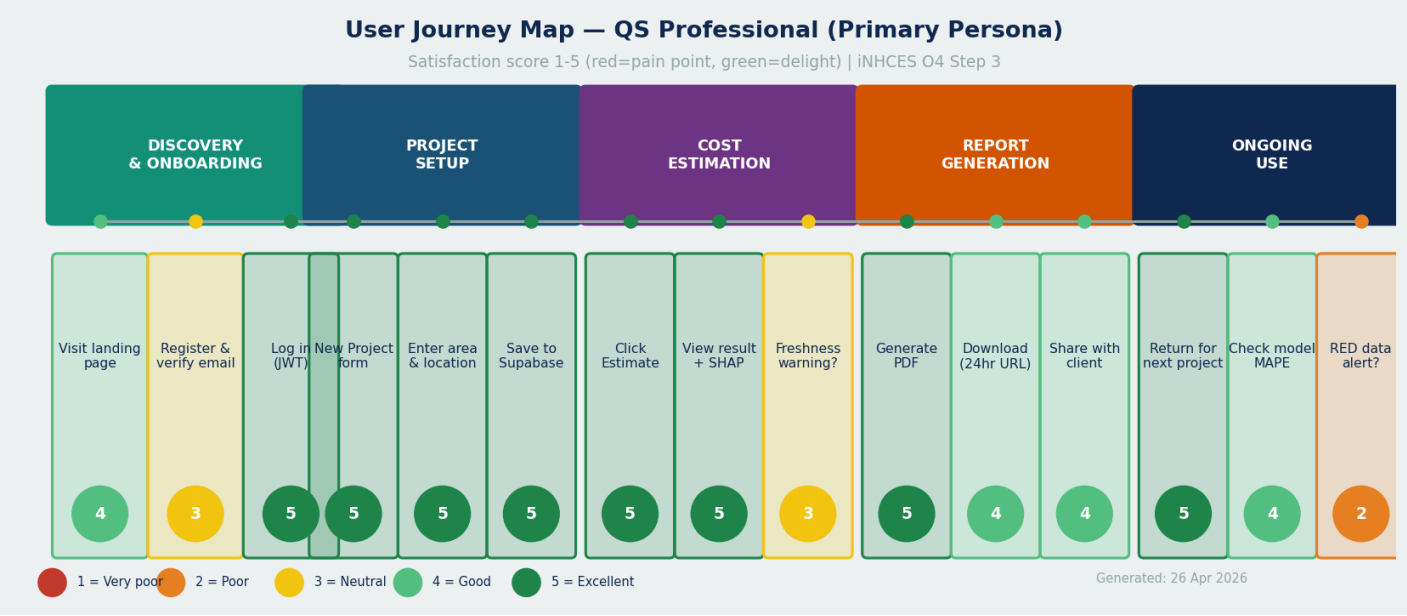
All 9 Airflow DAGs with cron schedules, external data source connections, Supabase table targets, ML retrain pipeline (6-task sequence), drift detection, and FastAPI inference link.



DATA SOURCE: AMBER | diag_05_pipeline.png | iNHCES O4 | ABU Zaria / TETFund NRF 2025

Diagram 6 - User Journey Map

QS Professional primary persona: 5 stages, 15 touchpoints scored 1-5. Pain points highlighted in red/amber. Delight points in green.



DATA SOURCE: AMBER | diag_06_user_journey.png | iNHCES O4 | ABU Zaria / TETFund NRF 2025